Chapter 8

The methods and value of measuring choice processes in economics experiments

Isabelle Brocas[a], Colin Camerer[b], Juan D. Carrillo[a], and Ian Krajbich[c]
[a]*Department of Economics, University of Southern California, Los Angeles, CA, United States,* [b]*Division of Humanities and Social Sciences, California Institute of Technology, Pasadena, CA, United States,* [c]*Department of Psychology, University of California Los Angeles, Los Angeles, CA, United States*

Contents

1 Introduction 316
2 Simple measures of choice processes 319
2.1 Response times (RT) 319
2.2 Mouse tracking 327
2.3 Mouselab (eye-tracking on a budget) 330
2.4 Eye tracking 335
2.5 Language and voice 338
3 Complex measures of choice processes: psychophysiology and brain activity (fMRI, EEG) 339
3.1 Psychophysiology and emotion 339
3.2 Measuring brain activity 343
4 Causal manipulation of choice processes 346
4.1 Gaze and other process manipulations 347
4.2 Transcranial electric/magnetic stimulation 348
4.3 Stress induction 349
4.4 Pharmacological treatment 350
5 Conclusions and looking forward 352
5.1 Results of a *de facto* focus group 354
5.2 Specific research topics 355
References 359

Abstract
There are many methods to record behavioral and biological measures of choice processes in experimental economics. These measures are widely used in experimental neuroscience and psychology, but are more rarely used in experimental economics. The simplest methods are response times, recording of computer mouse and eye movements (even web-based), and psychophysiological measures like skin conductance and heart rate, which can be recorded by small devices placed or worn on the body. There are also more expensive and complex data such as measures and manipulations of brain activity. This chapter explains details of how the simplest methods work, and highlights examples of how new laboratory process data can help answer fundamental questions in economics. Process measures can also be increasingly helpful as experimental and

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behavioral economists study a wider range of cognitive constructs, such as imperfect perception, salience, and attention. These constructs are often treated as latent variables inferable from choice, but can often be easily measured directly from process data.

1 Introduction

In most economics experiments, in the lab and particularly in the field, the only data that are recorded, stored, analyzed, and reported, are the choices that people make. Some experiments also elicit beliefs (Camerer et al., 2002; Nyarko and Schotter, 2002; Schotter and Trevino, 2014), elicit self-reports, or ask survey questions.

In this chapter we describe how many other aspects of choice processing can also be recorded, some very cheaply and others with more effort or machinery. These process data can help refine which descriptive theories are most accurate and can often invite special kinds of causal manipulations.

Choice process data is created between the time when a computer screen first pops up, or an experimental question is posed, and the time when a choice button is pressed or an answer is given, any feedback is presented, and an experimental trial effectively ends. (We will use "choice processes" synonymously with "procedures" and "algorithms".)

There are lots of methods for measuring the choice process, ranging from automatically-recorded response times (RT) to brain imaging tools like functional magnetic resonance imaging (fMRI). Many of these methods have low marginal costs, are not too difficult to learn (they are used by thousands of sophomore-level undergraduate research assistants), and are highly reliable when analyzed properly.

There are several possible advantages to using process data:

1. As behavioral economists become interested in a wider range of cognitive processes, these new theories will often make joint predictions about cognitive processes and choice data (e.g. as in level-k theories of strategic thinking). In other cases, choice-process predictions can be teased out from those theories as reasonable extrapolations to what values process variables would take during choice.[1]

[1] When process predictions are not plausible extrapolations from theory, data can be collected in which subjects are *instructed* to use a specified choice process (an "assigned procedure" paradigm), and their choice process can be measured. Data from their procedural process can then be used as a template to classify whether uninstructed subjects use similar processes. When the instructed process data from process P and uninstructed process data are similar, one can conclude that the uninstructed subject was using process P. The pioneering example is Bettman et al. (1990). Later examples are Sanjurjo (2015), and Oprea (2020). This is a very useful method for establishing empirical markers of process and has been woefully underused. Note, however, that process statistics from use of a trained rule will likely differ at least a little, from statistics resulting from a personal rule 'choice'. For example, a person assigned to maximize expected value might have slower RTs than a person who chooses that rule (because they are presumably familiar with how to apply the rule quickly).

In our view, economists are very good at hypothesizing multiple plausible theories for a small set of regularities. For example, think of the many theories of endowment effects, the Ellsberg paradox, and ultimatum game rejections as examples. Those theories often chase only a few data. One way to make better progress is to collect more types of data, and discriminate among theories on the basis of *both* choice and process data.

2. Most of the process measures we will discuss have been used in cognitive psychology for many decades. Therefore, economists do not have to start from scratch in learning what to do. Even electroencephalography (EEG) goes back more than 150 years, to 1875. Because of this long history of using process methods in psychology, there are a lot of published "users' manuals", tutorial articles, exemplar studies, and well-known tricks about how to collect good-quality data. In some of the authors' labs, high school STEM students have used some of the simpler methods we describe in this chapter (RT, eye-tracking and Empatica sensors). Every economics Ph.D. student can learn to use these methods.
3. Methods and devices for recording process data are getting cheaper and better. RT are essentially costless to measure. Webcam-based eye-tracking methods such as Webgazer (which is free at the margin), were introduced in 2009 and are promising for eye-tracking of online subjects. Two of our groups have used them for online-subject experiments; they are adequate to detect attention to six regions on a screen but not as good for finer spatial resolution. Good quality in-lab eye-trackers and mobile eye-trackers (wearable glasses) are now in the $500–$5000 range. Affordable wrist-worn (Empatica, Fitbit) and ring (Oura) sensors can measure a lot of psychophysiological data. The only method with a high marginal cost is neuroimaging.
4. The use of choice process data in economics is increasing rapidly (as in the early phase of an innovation diffusion process). Over the 2008 to 2023 period, the number of economics articles referencing choice process data has increased dramatically: 9,380 to 19,500 for response times articles, 747 to 9,500 for eye-tracking articles and 16 to 266 for mouse-tracking articles.[2]

In establishing the scientific value of process data for economics the crucial question is, and will still be in the foreseeable future: What interesting economic question can we answer *better* by including process data? Here are some early examples, most more than ten years old:

Economic question: How do people form beliefs in a "restless" bandit binary choice task? Hu et al. (2013) used eye lookup data as a proxy for beliefs about which of two bandit choices was more profitable. Beliefs inferred this way show deviations from Bayesian updating and more updating in response to loss than gain.

[2] These numbers are based on Google Scholar searches for articles published in those years.

Economic question: Are overly fair bargaining offers due to equilibrium reasoning including prosociality, or to non-equilibrium limits on human backward induction? Process data from Mouselab showed that in multiperiod bargaining, most subjects did not look ahead to future potential payoffs (which is necessary for equilibrium computation) (see Camerer et al., 1994; Johnson et al., 2002).

Economic question: Is trading too quickly in a "clock game" (Brunnermeier and Morgan, 2010), which was designed to understand price bubbles, due to physical anxiety? Kang and Camerer (2018) measured skin conductance response (SCR) during the game (a common measure of anxiety and arousal). They found that higher SCR was correlated with trading too early. As traders converged to equilibrium over time, showing more patience in trading, their SCR fell.

Economic question: Is temporal discounting associated with self-control? Brain imaging research on dieting and food choice implicates a region of the dorsolateral prefrontal cortex (DLPFC) (Hare et al., 2008) as active in implementing self-control. This led Figner et al. (2010) to disrupt activity in the DLPFC in a common smaller-sooner larger-later monetary choice paradigm that measures discounting. The DLPFC disruption led people to behave more impatiently. Together these two studies suggest self-control for both food and money share neural circuitry, and that expressed behavioral discounting can be causally changed by manipulating brain activity.

Economic question: Can correlates of private information that aren't manipulable be used to improve efficiency in public goods mechanisms? Krajbich et al. (2009) showed that it can work, as the theory prescribed, using neural measures of private values. This is an extension of the dictum "Don't ask the person, ask the brain" when a person may not know, or won't say, what mental process created their decision. Their method, using both stated valuation and neural measures, and a penalty when those measures mismatched, improved allocative efficiency.

Outside of the narrow confines and short time windows of experiments, there are three other reasons choice process measures can be useful.

Reason one: Behavioral economics often studies systematic "mistakes". Mistakes include incorrectly computing compound interest or only paying attention to the first digit of a price. These mistakes might reflect rational inattention, frictions, etc., but they are still mistakes that have harmful consequences. A plausible welfare criterion is therefore that correcting such mistakes at low cost, without creating other harms, could improve welfare (see Bernheim and Rangel, 2007, 2009, for short and long versions consistent with this argument). Such low-cost correction of common mistakes is often the goal of "nudging". Choice process data can often be crucial in detecting and correcting mistakes in this way. For example, if a webcam- or phone-based eye-tracker reveals that new

banking customers tend not to look at a small box showing how much their debt can grow if they only make the minimum payment, that is useful information for customers and regulators to know.

Reason two: In almost all online and lab economics experiments, people make choices rather rapidly, within a limited time on the order of minutes or hours. But a lot of consequential human choices stretch out over long periods: Think about choosing a school, choosing whom and when to marry, or buying a house. In these choices a large amount of process data can be collected (and experimentally treated) before observing a single choice. Extensive pre-choice process measures could help predict what people will eventually do, and also prevent possible mistakes (as described in the last paragraph).

Reason three: choice process data is often naturally observable in economic settings. The time that it takes someone to respond to an offer; the items that someone looks at while shopping; the physical hesitation that someone displays when making a purchase—these are all choice process measures that some agents display and that other agents may observe and use strategically. By ignoring these signals, we ignore relevant market information.

This chapter will talk about different ways to measure aspects of choice processes. Table 1 presents an overview. This should be seen as a quick reference for the rest of the chapter and not something to try to digest right away. The methods are roughly ordered from simplest (section 2) to most complex (section 3), to methods that allow for causal manipulation (section 4).

An important omitted category is using language data such as self reports and communication. Brandts et al. (2019) and Agranov (Chapter 4) cover aspects of this important type of data.

2 Simple measures of choice processes

2.1 Response times (RT)

We'll start with the very simplest measure that is also cheap to collect—response time (RT), also commonly referred to as reaction time or decision time. RT is simply the elapsed time between when a question is first posed and when the decision-maker responds. It is a natural by-product of decision making and can be recorded in nearly any setting.

As more economic transactions occur digitally, RT is becoming a standard variable in digital datasets too. In this section we discuss what can be learned from RT and how to design experiments to measure it accurately.

Psychologists have been studying RT since the 1800s. Cognitive psychology has been using it as a formal tool since the 1950s, to study attention, memory, perception, and decision making. Consequently, there is a lot that we know about RT—more than can be carefully summarized in this chapter. Instead we will highlight some stylized facts about RT that could be useful to experimental economists. Spiliopoulos and Ortmann (2018) wrote an excellent deeper review.

Here are some basic properties of RTs which are well-known:

TABLE 1

Method	What's easy and useful?	What's challenging?	What's most difficult?	Example 1	Example 2
Simple methods that can be used in standard experimental labs or online					
response times (RT)	easy to record and scale, lab or online	aggregates processing steps	mapping process to RT	Nielsen and Rigotti (2022). Longer RTs are associated with "I don't know" preference incompleteness.	Konovalov and Krajbich (2019). Slower decisions are more likely to be reversed later.
mouse-tracking	easy to record and scale, lab or online	easiest for binary choice, screen layout	multitude of trajectory measures	Stillman et al. (2020). Shorter mouse trajectories are associated with stronger risk preferences.	Sullivan et al. (2015). Mouse-tracking in food choice shows that taste is processed 195 ms before health. Health-taste processing times account for 15–40% of self control behavior.
Mouselab	easy to record and scale, lab or online	can promote overly systematic search	mapping process to Mouselab lookups	Camerer et al. (1994) (cf. Johnson et al., 2002). In bargaining, fair offers are associated with limited attention to future-payoffs.	Brocas et al. (2014). Subjects in betting games often overbet because they do not look up opponent-payoff information to infer what opponents will do.
eye-tracking	works best in-lab, but available online	screen layout, "calibration", difficult to scale without going online	mapping process to eye tracking statistics	Krajbich et al. (2010). Dwell time predicts choice according to a model of attention and DDM ("aDDM").	Polonio et al. (2015). In 2x2 games, classification based on gaze patterns predicts strategic sophistication and social preference.
voice	easy to record and scale, lab or online	requires machine learning tools	too new to know	Capra et al. (2023). Predicts cooperation and deception from voice features.	

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TABLE 1 (*continued*)

Method	What's easy and useful?	What's challenging?	What's most difficult?	Example 1	Example 2
Biophysical methods that require specialized equipment					
psychophysiology (e.g., skin conductance response (SCR))	measures arousal	costly information preprocessing	mapping measures to processes	Brocas et al. (2021). SCR predicts response to public "shaming".	Kang and Camerer (2018). Anxiety→ SCR→ different behavior in sequential vs reduced-form clock games.
EEG	high temporal resolution	fixed cost of equipment ($20-150k, some portable ones <$1000), limited depth below cortical surface (poor spatial resolution)	preprocessing, source localization	Polania et al. (2014). Frontal-parietal synchronization is associated with choice accuracy.	Harris et al. (2011). Value is constructed in reward regions.
fMRI	whole-brain coverage, high spatial resolution, wealth of existing data	fixed cost of equipment ($1-3M 3T), technicians needed, high marginal cost ($500/hr), head motion harms data quality, preprocessing, correcting for false positives	careful design to associate brain regions 1-1 with functional circuitry	Hsu et al. (2005). Ambiguity-aversion is associated with amygdala and lateral frontal cortex (LPFC) activity. Patients with LPFC damage are ambiguity-neutral.	Delgado et al. (2008). Overbidding is associated with loss-related activity in ventral striatum. A followup showed that emphasizing losses increased overbidding.

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TABLE 1 (*continued*)

Method	What's easy and useful?	What's challenging?	What's most difficult?	Example 1	Example 2
			Selected causal methods		
exogeneous visual exposure	tests the effect of attention on choice	details of exposure can matter	changing attention without changing the options	Pärnamets et al. (2015). Uses gaze manipulation to alter moral decisions.	Armel et al. (2008). Uses exposure times to alter consumer choices.
exogenous stress	reliable, IRB-approved, helpful to measure cortisol	Need to send sample to a laboratory to analyze data	heterogeneity in stress responses	(Lighthall et al., 2009). Stress increases risk-taking among men but decreases it among women.	Brocas et al. (2017). Stress reduces the ability to react to contextual information about games.
psychopharmacology	careful treatment, as with medical patients	precise timing if dose effect, degradation and spillover, differential dose-response, multi-chemical interactions	When is pharma expected to have a behavioral effect?	Brunnlieb et al. (2016). Vasopressin increases efficient coordination in stage hunt games.	(Declerck et al., 2020). Fails to replicate an effect of oxytocin on trust.
TMS	causes or inhibits neural firing	requires precise spatial placement based on MRI, can cause seizures in people with epilepsy, difficult to reach brain regions beyond the cortical surface.	IRB-approval	(Krajbich et al., 2021). Disruption of attention-related activity alters the effect of gaze on choice.	Baumgartner et al. (2011). Disruption of DLPFC activity alters Ultimatum game responses.
tCDS	upregulates or downregulates neural firing, can be run in parallel, easy sham (control) condition	small effects, difficult to target specific regions or penetrate deep into the brain, can be dangerous if improperly used	large samples required, attributing effects to particular brain regions	Li et al. (2020). Upregulation of rTPJ causes more prosociality and consistency with rationality.	Maréchal et al. (2017). Upregulation of rDLPFC reduces cheating in die-rolling by half.

First, RTs decrease with experience with a task (Logan, 1992). In experimental economics these decreases come from learning to use an interface, basic information processing, learning what others seem to do, and so forth (Dutilh et al., 2019). Fortunately, a logarithmic transformation of RT is often close to linear in periods or clock time. The relation is lawful so detrending works well to eliminate learning effects if they are a nuisance.

Second, RTs reflect a speed-accuracy tradeoff: People make more errors under time constraints (Ratcliff and McKoon, 2008). Time constraints can also interact with expectations or prior information to affect choice in different ways. Without any prior expectations, time pressure will induce more random responses and errors. But if prior expectations favor option A, for example, then time pressure will increase the choice bias in favor of A (Chen and Krajbich, 2018). This principle is illustrated in hider-seeker games with visually salient "focal points" (Li and Camerer, 2022). Both hiders and seekers automatically start looking at salient image features, although the top-down goal of hiders is to avoid choosing the same features as seekers do. Under exogenous time pressure, hiders lose much more often because the initial-fixation bias toward salience handicaps them. Drift-diffusion models (DDM) are a good framework to understand the speed-accuracy tradeoff and the influence of prior bias (modeled as a starting point bias, see below). Another illustration is from Spiliopoulos et al. (2018), who show that time pressure reduces complexity of strategies in games as measured by elementary information processes (a la Payne et al., 1993).

Third, RTs lawfully increase in the number of options or processing steps. It takes time for the brain to do additional steps of thinking (Gill and Prowse, 2023). In some cases a lot is known about the typical range of times that different "elementary information process" steps take (typically 50-500 ms). For example, each piece of information that a decision-maker has to look at requires a 120–200 ms fixation (see eye-tracking section below). An established principle called Hick's Law is the empirical regularity that RT for decision among N options increases logarithmically with N (Proctor and Schneider, 2018). Like some other "laws" in psychophysics, these relations are truly lawful within a data set because they have very high R^2 and are highly replicable in form (although parameter values differ).

It's important for experimental economics to know that RTs are also closely associated with fundamental economic concepts.

RT reflects strength-of-preference: Consider a choice between two options A and B, in which a decision-maker assigns values $v_A > v_B$ to these two options. Conditional on choosing A, the decision-maker's mean RT will be decreasing in $|v_A - v_B|$, that is, decisions will be faster when the absolute difference in values is larger (i.e. strength-of-preference, see Krajbich et al., 2014). The importance of this fact for economics is clear—utility differences can be inferred directly (Alós-Ferrer et al., 2021a,b), or estimated more precisely (Webb, 2019), if RT data are included. The Schotter-Trevino experiment on "global games" detailed

below is a good example. RTs can even allow us to predict choices that have never been observed. For example, suppose B is quickly chosen over A, but C is slowly chosen over A. If the quick and slow RTs are measures of utility differences (which they are) we can then predict that B will be chosen over C (Clithero, 2018), and C choices will be reversed more often in test-retest comparisons.

Small errors are slow, large errors are fast. Consider again the example above with two options and $v_A > v_B$. In these cases, sometimes B is chosen ($v_B \succ v_A$). Empirically, if v_A is only "a bit" larger than v_B ($v_A > v_B$), then these small errors are generally slow. Specifically, $RT(v_B \succ v_A) > RT(v_A \succ v_B)$. However if v_A is much larger than v_B ($v_A >> v_B$), then these larger errors are generally fast and the response time result reverses, $RT(v_B \succ v_A) < RT(v_A \succ v_B)$. Because small errors are far more common than large errors, slow responses tend to predict future choice reversals (Konovalov and Krajbich, 2019). We discuss some models of these phenomena below.

RT is well described by sequential sampling models such as the well-known drift-diffusion model (DDM): Many of these patterns in RT can be explained by sequential sampling "evidence accumulation" models which posit that decision-makers accumulate and compare noisy evidence in favor of each option until the relative evidence for one option over another reaches a pre-specified threshold. These models first emerged in statistics (Wald and Wolfowitz, 1948) and have been developed as models of human cognition ever since. Neural evidence for components of DDM were discovered in the early 1990s (Shadlen and Newsome, 1996) and later explored to explain economic decision-making (Krajbich et al., 2010; Brocas, 2012; Clithero, 2018; Fudenberg et al., 2018, 2020; Webb, 2019; Baldassi et al., 2020).

These sampling models all predict a joint distribution of choice probabilities and RTs, and RT distributions of errors (when errors can be clearly identified as in perception). A well-known special case is the DDM which yields logit choice probabilities and a skewed distribution of RT. This class of sequential sampling models has been shown to imply a random utility representation and can therefore be used to infer preferences from the combination of choice and RT data, yielding estimates of preference parameters such as the marginal effects alluded to by Bernheim (2009) with lower variance than can be achieved from choice data alone (Clithero, 2018; Webb, 2019; Konovalov and Krajbich, 2019). These models accommodate expectation effects by adjusting the starting point of the drift process, and accommodate the speed-accuracy tradeoff by adjusting the decision thresholds (Fig. 1). They also naturally produce the relationship between strength-of-preference and RT.

While the simplest version of the DDM predicts that the RT distributions, conditional on outcome (error or not), should be identical, the patterns of slow and fast errors that are typically observed can be modeled by the DDM with

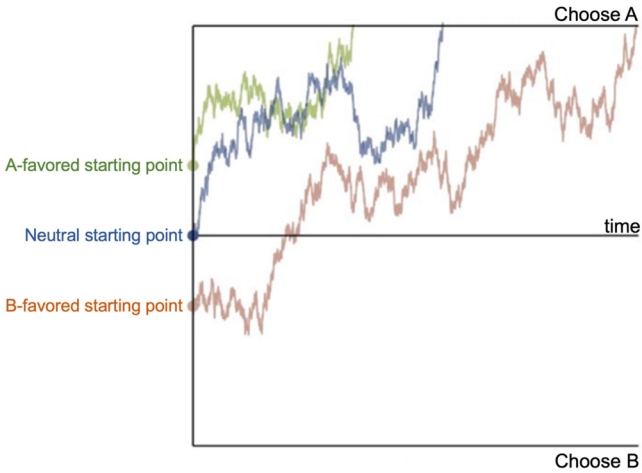

FIGURE 1 This illustration represents a typical DDM process with three different starting points. The y-axis is an accumulation of subjective value which 'drifts' linearly, but with noise, toward the upper threshold ("Choose A") or the lower threshold ("Choose B"). When accumulated value crosses a threshold a decision is made. The accumulation can start at different points depending on prior information or context. A starting point that favors Option A will increase the probability of choosing A and decrease the time it takes to choose A (but increase the time it takes to choose B). Source: Original artwork adapted from Izakson et al., 2024.

across-trial variability or, to a lesser extent, collapsing boundaries (Ratcliff and McKoon, 1997; Brocas, 2012; Fudenberg et al., 2018). Typically, these sampling models can be rationalized by standard models of accumulation of costly information. Bounds and speed of convergence to the bounds depend on primitive parameters such as the precision of the information accumulated, the size of the rewards when reaching a bound, and the cost of time. At each point in time, bounds are given by the optimal stopping rule of this model. Parallel bounds correspond to the solution of this problem without time limits. Collapsing bounds are predicted when the time by which a decision should be made is finite (see Fudenberg et al., 2020 for an econometric treatment). Finally, sequential sampling models emphasize that behavior is a joint distribution of choices and RT, therefore both are endogenous outcomes of the choice process. Thus they offer a theoretically-grounded model for understanding stochastic choice and the link between choice, RT, and exogenous observables, such as those decided by the experimenter.

How to measure RT: Most modern software packages will record timestamps with precision at the millisecond level, although researchers should be wary of lags that can be present when participants are online. In online field settings, users may disengage from the decision task for long periods. For this reason it is particularly important to do some data cleaning, or if not, to only analyze RT in settings where subjects make relatively quick decisions.

From a DDM point of view, the total RT is an aggregation of processing steps that are useful to distinguish. It is important to distinguish thinking about options to make a decision (drifting or accumulating evidence) and all other "non-decision time". Non-decision time is time before decision (e.g. reading instructions and identifying options) and after decision (e.g. pushing a button). It is important to control or measure non-decision in order to unpack total RT.

When analyzing RT data there are a few pre-processing steps that you should take. You may want to screen out responses that are exceptionally fast or slow. Very fast responses may indicate random guessing, while slow responses may indicate inattentiveness. A common approach is to exclude a response if it is more than 1.5 times above the third quartile or below the first quartile of the RT distribution for that subject. When analyzing RT as the dependent variable, say in a regression, it is also common to log-transform the RTs so that they are roughly normally distributed. Note that valuable information about the subjects may still be obtained from these exceptionally fast or slow responses.

Exemplar studies: RTs have been advocated as a simple way to define "instinctive" and "contemplative" strategic decisions. As Rubinstein wrote (2016, p. 874) "By "instinctive" we mean that the subject follows a gut feeling without applying any strategic analysis. By "contemplative," we mean that the subject analyzes the strategic aspects of the game and bases his choice on what he expects the other players will do."

This distinction relies entirely on fast and slow responses. It's a start. But other research has argued, as discussed above, that inferring instinctive decision-making from fast responses is problematic because fast responses also result from strong preferences/beliefs (Krajbich et al., 2015).

Chen and Fischbacher (2016) found that RTs were slower in dictator allocation games when choice-classified "types" had a stronger conflict between selfish and social motives.

Schotter and Trevino (2021) is an exemplar study using RT in a sophisticated way. They exploit the property mentioned above, that when two choices are close together in value, RTs will be slow. They apply this principle to identify thresholds in a global game.

In their game, players simultaneously choose A or B. There is an unknown hidden variable θ with a known distribution N(50, 2500). Each of two players i get a conditionally independent signal x_i which is the true θ plus a small deviation (with variance equal to one). Then they play a simultaneous stag hunt game shown in Table 2. Choosing A gives $\theta - 18$ if the other player also chooses A and -18 otherwise. The B payoff is always zero. The participants play 50 rounds with different θ and signals each time.

It is well-known in these games that players should (in equilibrium) and do (in practice) use an approximate threshold strategy: They choose A if the signal x_i is above a threshold θ^* and choose B otherwise. The authors therefore use

TABLE 2 Global game structure (Schotter and Trevino, 2021).

	A	B
A	$\theta - 18$	-18
B	0	0

the trial with the slowest RT to estimate θ^*, on the presumption that a slow RT means the signal the subject received, which made them slow to respond, was probably close to the θ^* threshold. The two slowest average RTs were around 12 and 8 sec, which are much slower than the median of 3 sec. Inferring θ^* from the slowest RTs gives a very good prediction of the threshold estimated statistically from the last 25 trials.

Konovalov and Krajbich (2023) push the scientific use of RT even further in a bargaining domain. They find that after a fast rejection (rapid RT) of a seller's offer, the seller infers that she is far from the buyer's reservation price and follows up with a larger price drop than if the buyer had rejected slowly. Moreover, when given a choice to reject a seller's offer quickly or slowly, the buyer chooses to reject quickly, correctly intuiting that this should yield a lower second offer.

A "many analysts" study of RT by Dutilh et al. (2019) is a good illustration about drawing inferences about latent process models from RT. They assembled 14 random dot motion data sets with three separate treatment conditions. Fourteen teams, blinded to the exact treatment differences, were tasked with testing for differences using whatever form of DDM model they chose. The idea is to see how much "researcher degrees of freedom" changes conclusions on a fixed set of data. Results were mostly overlapping but there were some differences. Happily, simpler methods seem to identify treatment effects about as well as more complicated ones.

2.2 Mouse tracking

Mouse tracking is a technique that can allow us to infer aspects of the decision-maker's thought process as their decision unfolds, by recording the position of the mouse cursor as it moves around the screen. It is a relatively recent technology (c 20 years old), facilitated by the widespread adoption of computerized experiments (Spivey et al., 2005). Since people naturally try to maximize the efficiency of their hand movements, they tend to move their mouse more rapidly and linearly towards where they expect to end up. Like RT, mouse-tracking data is becoming more available and relevant for economic transactions, which are increasingly occurring online.

In a typical mouse-tracking experiment, the subject initiates a trial by clicking a "start" button somewhere on the screen. They then move the mouse to one of several choice options. Mouse-tracking records and analyzes the path that the mouse cursor takes as it travels from the start to end point.

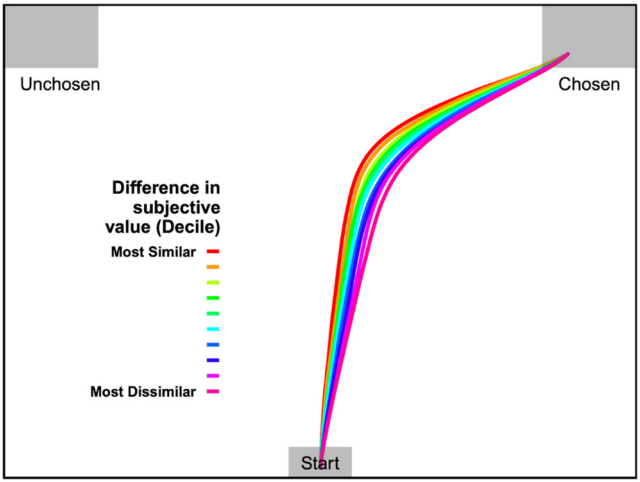

FIGURE 2 Mousetracked trajectories reflect strength of preference, as measured by the difference in subjective value between the chosen and unchosen option. Each line is an average of trials binned into ten deciles. The most dissimilar values (the largest gap in value for the chosen option) lead to the most direct trajectories. Source: Stillman et al., 2020.

Mouse tracking reflects strength-of-preference: Generally speaking, more direct trajectories are associated with stronger preferences (Fig. 2). This can be established using various statistics (see below) such as the area under the curve.

Changes in mouse trajectories over time reflect attribute latencies: In multi-attribute decisions, we can use mouse trajectories to infer the average latency (i.e., delay) with which each attribute begins to influence the decision (Sullivan et al., 2015; Fisher, 2023; Chen et al., 2024a). Using a time-series regression, the first time at which each attribute has a significant effect on the location of the mouse cursor can be estimated (Fisher, 2023). Attributes with shorter latencies tend to have more influence on choice outcomes and are predictive of how choices will change under time pressure.

Mouse trajectories can reveal subjects' beliefs: In the game of "whack-a-mole" there are holes on a screen and moles pop up periodically from different holes. The goal is to hit the moles as soon as you see them. This is a useful paradigm for learning patterns if there is some pattern in where the moles pop up.

Imagine playing whack-a-mole and detecting a pattern in the moles' locations. You would begin to anticipate the location of the next mole and move your mouse in that direction early in the process. This early mouse movement is a measure of a belief about where the mole will pop up next. Konovalov and Krajbich (2020) studied this phenomenon in a learning task where subjects clicked one of two options at the top of the screen and then had to move their mouse to the bottom of the screen to claim their reward. After a delay, the reward showed up either at the bottom-left or bottom-right corner of the screen

and the location was predictable based on the chosen option. Subjects' mouse movements revealed that they learned these patterns; after clicking an option at the top of the screen, their mouse cursors began to move towards the more probable corner. The movements were larger for more extreme probabilities.

This same approach could be used to study subjects' beliefs under uncertainty, in social dilemmas, etc.

How to measure mouse location?

You can record mouse location using most software packages, including Javascript (jspsych, Qualtrics), Python (psychopy, oTree), Matlab® (psychtoolbox). There is also popular free software for mouse-tracking (https://www.mousetracker.org/user-base). These tools will record mouse cursor location (x and y coordinates) every few milliseconds. You can expect measurements approximately 50 times per second, which is a high enough sampling rate for most decision-related tasks.

Requiring continuous mouse movements

One important methodological detail is whether to require subjects to keep their mouse moving throughout the decision. Some researchers insist that subjects keep their mouse moving once they click the "start" button and the options are revealed. This is to prevent subjects from making their decision and then ballistically moving their mouse directly to their chosen option. However, this approach has some drawbacks. It is less natural and requires informing subjects that their mouse movements are being recorded. If a subject needs a lot of time to decide, this might lead to artificial behavior like moving the mouse in a circle while deliberating. This approach also precludes studying idle vs. motion time. An alternative approach is to simply let subjects move the mouse unconstrained. While this might lead to straighter trajectories, it can also work effectively (Cheng and González-Vallejo, 2017).

Mouse hardware

A second issue is whether subjects are using a mouse, trackpad, or trackball. These different hardware options will lead to somewhat different trajectories. For example, a trackpad will lead to straighter trajectories than a traditional mouse. For within-subject comparisons this is less of a concern, but for between-subject comparisons it is worth controlling for the hardware, if possible. For online experiments, this can be self-reported or sometimes recorded (e.g. using JavaScript code).

Task design

The typical mouse-tracking experiment involves binary choice, with the "start" button at the bottom center of the screen and choice options located at the top-left and top-right corners of the screen. For multiple options (or options on a

continuum—e.g. willingness-to-pay), the start button can be located at the center of the screen, with the choice options located in a semi-circle around the choice option; we use a semi-circle to fix the distance between the start button and every possible choice.

Mouse-tracking statistics: Different statistics have been used to measure how direct the path is that the mouse cursor takes from the start to end point. The most popular is "area under the curve" (AUC) (also called "absolute average deviation" or AAD). This is the gap between measured trajectory and a straight line connecting the start and end points. If the mouse cursor traveled linearly from start to end, AUC = 0. The less direct the trajectory, the larger the AUC.

This statistic is similar to, and correlates strongly with, RT. Like RT, AUC correlates negatively with strength-or-preference; we observe larger AUC for choices between options that are closer in subjective value (utility). AUC sometimes has an advantage over RT in that it is less likely to be contaminated by non-decision components (e.g. Stillman et al., 2020 found correlations with risky-choice parameters of $r = 0.25$ with AUC and $r = 0.17$ with RT).

Other statistics include the maximum deviation from linear, X-flips (direction changes from heading left to heading right and vice versa), total distance, idle vs. mouse motion time (which may reflect deliberation that pauses mouse movement), and acceleration.

2.3 Mouselab (eye-tracking on a budget)

A time-tested way to measure choice processes, in the form of sequential search for information, is to hide information behind opaque boxes in labeled locations and measure explicit physical actions which 'search' for that information. The modern software-based version of this paradigm is called "Mouselab".

It is crucial to note at the start that the interpretation of Mouselab and camera-based eye-tracking data as measures of latent cognitive process rely on assumptions. In the context of reading, Just and Carpenter (1980, p. 330) described the "eye-mind" hypothesis as "the eye remains fixated on a word as long as the word is being processed".

One purpose of visual fixation is undoubtedly to scoop up information to hold in working memory to eventually compare with new information. But there will be temporal decays in this memory, and biased future attention based on what has just been seen (the brain is always predicting). Refixations—looking back at an object again—are also interesting: Are they refreshing memory or accumulating more evidence a la aDDM?

Sophisticated econometrics might enable experimental economists to do a little better than other disciplines in describing the mapping from fixations to choices, especially for more complex problems than choosing between two objects.

In the original process-studying paradigm of this sort, Payne (1976) created a physical information board in which different valued attributes of apartments

were organized in a matrix (apartments were rows, and attributes such as rent and square footage were columns). The actual attribute values were hidden behind wooden tabs which could be physically lifted up to show their values. Payne, Bettman, Johnson and colleagues later moved to a "Mouselab" paradigm in which the variables hidden in boxes that were also labeled (such as probabilities and possible outcome values), and the values were revealed when a mouse moved into the box area, which 'opened up' to show the searched-for value. (Think of the children's game "Memory" where tiles are flipped over to reveal items underneath.) A careful decades-long research program showed that people do not lift up all the tabs (in the 1974 study) or open all the boxes (summarized in Payne et al., 1993). Instead, people exhibit a heuristic, roughly efficient search for a small, adequate set of diagnostic attributes.

This kind of design can feel unnatural, but can be very useful for studying attention because it allows the researcher to observe exactly what information subjects have when they make their decision and in what order it was collected. Because humans can attend to information even if they are not looking directly at it, hiding the information that is not being pointed to eliminates this potential confound. However, there is some evidence that Mouselab encourages more systematic search processes (Lohse and Johnson, 1996). See the section below on "gaze contingent eye-tracking" for related information.

The Mouselab paradigm has been widely used in marketing and psychology to understand what economists often call "procedural rationality"—i.e., is there an identifiable procedure being used in response to task demands? Those experimental economists who lean toward understanding pure cognition could find Mouselab type measures useful too, to the extent that they are pursuing the Payne et al. (1993) procedural rationality research program.

Mouselab and eye-tracking have proved especially useful in three domains of economic interest: Multiattribute choice (the Payne-Bettman-Johnson program described above); risky choice (e.g., Harrison and Swarthout, 2019); and game theory.

Mouselab and eyetracking are especially useful when there is an hypothesis that some players form incorrect beliefs because they do not attend to all elements of the game structure.

Games are routinely displayed visually as matrices of payoffs or trees. Many game-theoretic equilibrium concepts, such as backward induction or iterated dominance, require a specific way of looking at, and comparing, payoffs if they are efficiently implemented by eye-hand coordination. Visual lookups are therefore an ideal dataset, along with choices, to test competing game theory ideas.

In such scenarios, behavior can be rationalized by various models, such as Cognitive Hierarchy and level-k models (e.g., Camerer et al., 2004; Crawford, 2008), which can be tested by combining choice and attention data. For instance, level-k theory posits that if level 0 players randomize equally, then level 1 players will make choices that maximize expected payoffs given those beliefs. Level 1 players will not need to look at the other players' payoffs, although they must

still be aware of their opponents' possible actions over which they randomize. By checking whether subjects focus attention on these payoffs, researchers can gather evidence of level-1 reasoning.

With MouseLab, information is hidden behind blank boxes. In the early versions of MouseLab (Payne et al., 1993; Camerer et al., 1993; Johnson et al., 2002), subjects did not have to hold down a button to keep information visible. In later versions, information can be revealed by moving the mouse into the box and clicking (Costa-Gomes et al., 2001; Costa-Gomes and Crawford, 2007) or clicking-and-holding (Brocas et al., 2014, 2018a,b) the left button. If subjects release the button, the information disappears. The clicking-and-holding variant ensures that subjects are attending to the information and eliminates the need to filter out boxes opened for a short time (as defined by the experimenter). It also ensures that an unusually long time spent in a box reflects a long fixation rather than the subject leaving the mouse in that position while daydreaming. There is no restriction on the amount, sequence, or duration of clicks, and no cost associated with it, except for the subject's effort, which is assumed to be negligible. The software records the sequence and duration of clicks.

MouseLab is a simple and cost-effective way to measure what information people might be paying attention to. It scales up cheaply because it can be used on many computers simultaneously. This scalability is advantageous for studying multi-person games and markets. Single-subject eye-tracking using camera-based systems has better resolution (and also records pupillometry, a useful measure of arousal and cognitive difficulty), but they are significantly more expensive per unit.

An early example of this use of Mouselab comes from the substantial evidence about the nature and causes of "preference reversals" (aka violations of procedure-invariance). A common type of reversal involves bets of the form (p, X). A "p-bet" has high p and low X. A "$-bet" has low p and high X. Preferences can be elicited through binary choice or by asking a person to price both bets separately (cf. Chapman and Fisher, this volume). Reversals are common, in which the p-bet is chosen more often but valued less than the $-bet.

Using Mouselab, Schkade and Johnson (1989) found that people looked more at the box containing the p-value during choice, compared to their lookups during pricing. They also found that when using a $ scale to choose prices, participants tended to anchor near the X value of the $-bet which partly explained reversals. An exogenous causal change in setting the starting point reduced reversals.

Another example of the inferential value of Mouselab is experiments on "betting games" with two-sided asymmetric information, analyzed in Brocas et al. (2014). Behavioral data have long shown that people often fail to correctly anticipate how others use their private information, leading to mispricing of items with uncertain values, trades that should not occur, or overbidding. The patterns in these data reveal regularities in reasoning, indicating that people use flawed algorithms that produce systematically incorrect outcomes. However, it

	A	B	C	S
1	25	5	20	**10**
2	**0**	30	5	

FIGURE 3 Representation of the Betting game. Cells represent the payoffs of player 1 and player 2 if they bet and the state turns out to be A, B or C, or if at least one of them does not bet, in which case both receive payoff S. Vertical lines identify the partitions of information. Source: Brocas et al. (2014).

is challenging to determine from behavioral data alone whether this is due to incorrect calculations, choice errors, or focusing on the wrong elements of the decision problem. Non-choice data can help distinguish between these hypotheses. To address this issue, Brocas et al. (2014) consider the following game.

There are three possible states of the world: A, B, and C, with one state randomly determined by nature. Two players simultaneously decide whether to bet or not, without knowing the actual state. Each player receives a private signal: the first player learns whether the state is A or not, and the second player learns whether the state is C or not. If the state is A or C, one player is fully informed. If the state is B, the first player knows it is not A, and the second knows it is not C. Players' payoffs if both bet depend on the actual state, as given by the matrix in Fig. 3. If at least one player does not bet, they each get the sure payoff S. Under this payoff structure, the (Trembling Hand) Bayesian Nash equilibrium is to never bet. However, we often observe significant betting, which suggests that players may not be processing the decision problem correctly.

To understand why, we shall first note that there are 3 levels of complexity in this game. For instance, if Player 2 knows that the state is A, they only need to compare their payoff if both bet (0) with the sure payoff (10). That is, they only need to pay attention to their own boxes [2A] and [S] before concluding that betting is suboptimal.

Now suppose that Player 1 knows that the state is not C. They must look at box [2A] to determine if Player 2 will bet if the state is A, and also examine their own possible payoffs in [1A] and [1B], as well as [S]. This process is strategic because it requires considering another player's payoff, understanding that the opponent has different information sets, and making an inference based on the simplest level of rationality (dominance) of the opponent. In this example, once Player 1 realizes that Player 2 will not bet under state A, she has no incentive to bet when the state is not C as the payoff in [1B] is lower than [S]. This situation is called D1.

Lastly, consider Player 2 when the state is not A. Knowing whether Player 1 will bet under C is not sufficient. She must deduce whether Player 1 will bet when the state is not C, which requires her to consider her own [2A] payoff. In sum, they need to examine all of their own payoffs [2A, 2B, 2C], as well as [1A, 1B], and [S]. Notice that looking at [2A] means considering a payoff they know with certainty will not occur (an impossible counterfactual). This is highly counterintuitive and a clear hallmark of strategic reasoning. This situation is called D2.

These three situations can be unambiguously categorized from simplest to most complex, based on the degree of strategic thinking required. By tracking the lookup patterns, duration in certain boxes and transitions from one box to another, the researcher can determine if the subject is paying attention to the information necessary to compute the Nash equilibrium and, if not, identify the alternative algorithms they use.

The challenge with Mouselab is the enormous amount of data it provides, which then needs to be classified and studied systematically. Costa-Gomes et al. (2001) developed a method to analyze attentional data by focusing on two main measures. First, they consider whether a particular box (e.g., one that needs to be opened by subjects with a certain behavior) has been looked at or not. This measure is labeled "occurrence." Second, they examine whether, after opening a box, the next one to be opened can be rationalized by a certain reasoning process. This measure is labeled "adjacency."

In Brocas et al. (2014), the method helps illustrate that people often focus on a subset of information, supporting the idea that attention is costly. The study also shows that correct information lookups are very significantly associated with Nash behavior, while specific incorrect lookups lead to particular mistakes. Notably, people often overlook the counterintuitive payoff in a state they know will not occur, even when this information is crucial. Occurrence and adjacency measures are different between participants who look at the correct information and play Nash, those who look at the correct information but do not play Nash and those who do not look or play correctly (see Fig. 4).

In another application of MouseLab, Brocas et al. (2018a) distinguish the thought processes in sequential v. simultaneous dominance solvable games. In their paper, illustrated in Fig. 5, four players choose between two actions each, and their payoff depends on their action and that of one other player. By applying iterated elimination of strictly dominated strategies, it becomes clear that the equilibrium actions for players 1, 2, 3, and 4 are X1, Y2, Y3, and X4, respectively (highlighted cells in Fig. 5).

However, the level of strategic sophistication increases monotonically from the rightmost to the leftmost player. Player 1 encounters a simple game with a dominant strategy, thereby requiring no strategic thinking. In contrast, Players 2, 3, and 4 must iteratively eliminate 1, 2, and 3 strategies of their rivals, respectively, to determine the equilibrium. Interestingly, both the game's solution and the algorithm needed to solve it remain the same whether the game is played simultaneously or sequentially (starting with player 4). Instead, behavioral data indicates that the simultaneous version of the game is more complex. By tracking how subjects navigate the game, researchers can pinpoint what makes the simultaneous version more challenging: difficulty to identify the starting point of the algorithm (player 1), difficulty to iterate once the starting point is identified, and/or use of heuristics and shortcuts that result in wrong decisions.

The study demonstrated that people look at information differently in the simultaneous and sequential versions of the game, even among those who play

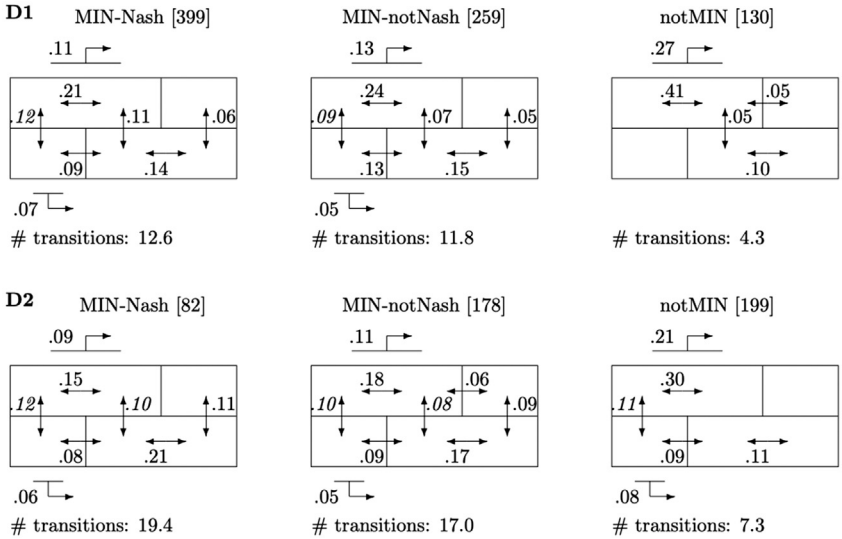

FIGURE 4 Transitions of lookups in D1 and D2 [# observations in brackets]. MIN refers to the minimum information necessary to look at in order to be able to find the Nash equilibrium. Representation is from the perspective of Player 1 in {A,B}. Number of transitions are highest among participants who play Nash and look at MIN (MIN-Nash). They are also higher in D2 compared to D1. Participants who do not look at MIN (notMIN) concentrate their lookups on the partition {A,B}. Participants who look at MIN and play Nash follow the algorithm that leads to Nash more closely. Source: Original artwork reporting new analysis of Brocas et al. (2014).

Payoff of 4

	X_3	Y_3
X_4	15	25
Y_4	30	14

Payoff of 3

	X_2	Y_2
X_3	38	18
Y_3	18	32

Payoff of 2

	X_1	Y_1
X_2	14	36
Y_2	30	10

Payoff of 1

	X_4	Y_4
X_1	34	26
Y_1	20	12

FIGURE 5 Representation of the game. Each matrix represents the payoffs of the row player if they and the column player choose their respective X and Y actions. Highlighted cells represent equilibrium actions (Brocas et al., 2018a).

according to Nash equilibrium. In the simultaneous version, there are more forward transitions. However, when participants begin to focus on the rightmost player's payoff matrix, they quickly start using backward reasoning to solve the problem. The sequential format provides a clue that must be discovered in the simultaneous format, making it more challenging to play according to Nash equilibrium in the latter case.

2.4 Eye tracking

Eye tracking allows researchers to monitor where a person is looking while making decisions. Initially used to study reading (Rayner, 1978), it was later adapted for multi-attribute decision-making (Russo and Rosen, 1975; Russo

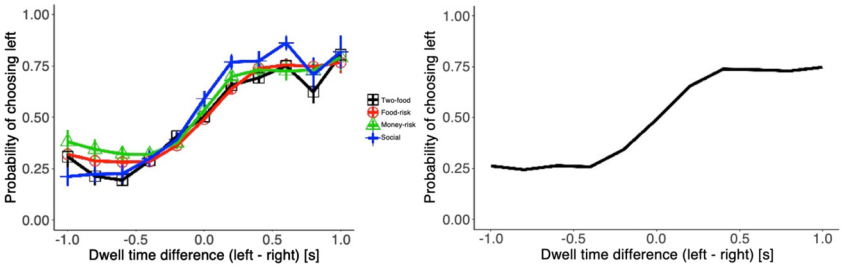

FIGURE 6 There is a robust relationship between relative looking time and choice, as seen in (left) binary choice tasks involving snack foods, monetary lotteries, food lotteries, and mini dictator games, and (right) simulations of the attentional DDM. Reproduced from Krajbich (2019).

and Dosher, 1983; Russo, 2011). People tend to look at what they are attending to (Schotter et al., 2010), making eye tracking useful to reveal information considered during decision-making. Importantly, gaze behavior is unaffected by being tracked (Kee et al., 2021).[3] Gaze movements are discrete, jumping between points in steps called "saccades" (20-40 ms), while pauses are "fixations" (several hundred ms). "Dwell time" refers to cumulative fixation duration within an area-of-interest (AOI), offering insights into choice behavior. Other measures like blink rate and "microsaccades" (short saccades which cannot be seen by low-resolution eye-trackers) are other useful measures. Wedel et al. (2023) is a good recent review of eye-tracking used to study decision making, particularly in psychology and marketing.

What have we learned from eye tracking?

Applications by economists often treat attention as binary (considered or not), but eye tracking provides finer-grained data. Covert attention allows information processing without fixation, e.g., peripheral vision avoids low-value items (Thomas et al., 2021). Gaze data can approximate attention, as illustrated by Polonio et al. (2015), where participants in matrix games focused only on their payoffs, missing their opponents' strategies.

Attention enhances processing. Dwell time correlates with choice probability (Krajbich et al., 2010) and attribute weight in utility functions (Fisher, 2021; Yang and Krajbich, 2023), a fact that is well captured by the attentional DDM (Fig. 6). This effect may amplify value-consistent information (Smith and Krajbich, 2019; Sepulveda et al., 2020; Westbrook et al., 2020) or be driven by other mechanisms (Callaway et al., 2021; Li and Ma, 2021; Jang et al., 2021). The meta-analysis by Bhatnagar and Orquin (2022) confirms that dwell time and last fixation affect choices, while first fixation might not.

[3] The term "lookup" is commonly used in experimental economics to refer to the presence of fixations and their duration". "Gaze" is a synonymous term used in psychology.

Comparing AOIs reveals strategies. In risky choices, gaze alternates between probabilities and outcomes, correlating with expected-value choices (He and Bhatia, 2023; Aimone et al., 2016). In other contexts, comparisons occur within attributes (Fellows, 2006; Noguchi and Stewart, 2014; Yang and Krajbich, 2023), influencing context effects (Landry and Webb, 2021). The Payne Index quantifies these patterns (Payne, 1976). The Payne Index is defined as the difference between within-alternative and within-attribute transitions, divided by the total number of transitions. A score of 1 indicates fully alternative-based search while a score of −1 indicates fully attribute-based search.

People don't look at all their options. They often skip low-value options in large choice sets, even without time constraints (Reutskaja et al., 2011; Thomas et al., 2021). This behavior is clear evidence that people only attend to subsets of a presented choice set—these subsets are called "consideration sets" in marketing and economics.

Visual salience affects gaze. Bottom-up salience depends on color, contrast, size, etc., while top-down salience directs attention toward goals. Neural networks predict bottom-up visual salience very well, even in "zero-shot" cases in which the stimuli were never seen in training (Li and Camerer, 2022). Combined with sophisticated methods for changing images, the ability to reliably predict eye gaze from algorithmic salience invites lots of interesting causal experiments, in which manipulated salience causes changes in gaze which potentially cause changes in choice.

How do we implement eye tracking?

Measuring eye tracking. Modern eye tracking uses high-speed cameras to capture gaze. Affordable options now provide sufficient spatial and temporal resolution for economics experiments, unlike high-end systems (e.g., EyeLink 1000 Plus) designed for psychophysics. Webcam-based systems (WebGazer; Yang and Krajbich, 2021) are improving but remain noisier. A tutorial on webcam-based eye-tracking (as well as mouse-tracking) is available at https://github.com/krajbichlab/sjdm_process_tracing_workshop. Mobile and virtual reality eye tracking show promise, though data analysis is more complex due to changing fields of view. One solution is to use controlled retail environments designed for research, where the researcher knows where each item is and can control these features.[4]

Calibration/Drift correction/AOIs separation. Calibration ensures gaze data accuracy by mapping eye images to screen positions. Drift correction is needed every 10–15 minutes to maintain precision. AOIs must be spatially distinct to avoid ambiguity, favoring circular designs to match error distributions. A central fixation cross should be added between decisions to ensure that participants start off each trial at a "neutral" location. It is important to randomize/counterbalance the locations of options and attributes across trials or participants.

[4] There are such labs at INSEAD, Duke, and Columbia.

Gaze-contingent design. These paradigms conceal information until fixation, eliminating peripheral vision effects. While less natural, they enable precise data on considered information, resembling MouseLab approaches.

Pupil dilation. Pupil dilation (PDR) reflects arousal and cognitive effort but requires careful control due to sensitivity to luminance and novelty. Baseline measurements are critical, and analyses must focus on specific time windows to capture dynamics. PDR correlates with risk (Preuschoff et al., 2011), the endowment effect (Sheng et al., 2023), loss aversion (Sheng et al., 2020), strategic misrepresentation (Wang et al., 2010), motivation (Alós-Ferrer et al., 2021a,b), and curiosity (Kang et al., 2009). More guidance on PDR analysis is available in Mahanama et al. (2022).

In summary, eye tracking provides rich, high-resolution data about attention and decision processes. It resembles and can be used to study similar questions than MouseLab, but it is certainly not identical. MouseLab is cheaper, slightly simpler for data analysis and more easily scalable, although the gap is closing thanks to the improvements in eye tracking hardware and software. Eye tracking is more accurate, non-intrusive and ecological, especially with the recent advances in equipment and techniques.

Advances in mobile tracking and machine learning for salience prediction promise broader applications, despite challenges with calibration, data interpretation, and integration with economic models.

2.5 Language and voice

Communication is an important part of economic activity, and particularly in interactive environments. Economics experiments have started using designs that allow participants to exchange information before making decisions.

Agranov (Chapter 4) discusses some aspects of communication in experimental economics. In this section, we briefly focus on a special aspect—information carried by audio properties of voice.

Voice can be collected and analyzed with simple audio recordings, so it can be done cheaply and non-intrusively. The ability to analyze voice data to measure emotions and intentions in real-time allows for immediate feedback and adjustments, which is particularly useful in dynamic environments like live negotiations or customer service interactions. Machine learning algorithms have revolutionized the analysis of voice data, enabling researchers to uncover subtle patterns and psychological markers that can reveal an individual's emotional state and intentions (Stoitsas et al., 2022; Vrij et al., 2000; Sondhi et al., 2016; Pérez-Rosas et al., 2015). Capra et al. (2023) provide an extensive description of the method that we briefly summarize here.

Preprocessing

Videos can be converted to audio WAV clips ensuring a frequency of 16,000 Hz, a bit rate of 128 kbps, in a mono channel format. Using Audacity (free software) audio clips can be cleaned to remove voices of no interest and reduce background noise. Clips are then normalized to consistent volume.

Extracting acoustic features

Free open-source software ***PRAAT*** can be used to analyze and manipulate phonetics and speech data (in a Python environment with Parselmouth). The next step involves extracting key acoustic features from the cleaned audio clips. Those features include pitch and pitch variation (standard deviation), harmonic-to-noise ratio ("roughness"), perturbation, and resonant peaks called "formants". Additional features, such as emotions and scripted speech, are also extracted. ***Empath*** is a voice emotion AI package to detect emotions (joy, calmness, anger, sorrow) from speech energy levels. Speech-to-text can be translated using IBM Speech, and analyzed for sentiment direction using the ***VADER*** tool.

Because vocal features, emotions and sentiment produce a large body of high-frequency data, it is often ideally analyzed by machine learning models which select predictive features by reducing many dimensions to guard against overfitting (implementing a bias-variance tradeoff). Common examples are decision trees and random forests.

In an exemplary study, Capra et al. (2023) analyze how acoustic features of speech can be used to identify cooperation and deception in high-stakes environments. They analyze voice recordings from two datasets—a TV show called "Golden Balls" which implements a prisoner's dilemma with preplay "cheap talk" for high financial stakes, and a set of courtroom testimonies. They found that vocal cues can reveal psychological states and intentions. For example, in their data higher pitch and lower intonation (pitch variation) correlate with deceptive speech, achieving a classification accuracy of 58% to 79%.

Voice analysis adds another dimension to simpler NLP analysis of text. There are potential applications in areas of experimental economics such as communication in bargaining, public goods games, asset markets, sender-receiver games and Bayesian persuasion.

3 Complex measures of choice processes: psychophysiology and brain activity (fMRI, EEG)

In this section we describe process measures which are more complicated to use, and require more difficult, expensive machinery. These methods also require "preprocessing", which means that raw recordings need to be transformed in some way before analysis. There are also specialized methods of analysis which are conventional wisdom among users of the methods. Note, however, that in many cases there are some differences in schools of thought about which measures to use, similar to ongoing econometric debate about details of causal IV and other methods.

3.1 Psychophysiology and emotion

The notion that economic decisions are influenced by emotions is intuitive and underpins various behavioral theories. Emotions can deter us from taking risks

and cause us to experience regret or guilt (Loomes and Sugden, 1982; Loewenstein et al., 2001). We may make tradeoffs to avoid negative emotions (Tangney, 1995; Eisenberg, 2014) such as guilt (Charness and Dufwenberg, 2006). Emotions can also serve as incentives; for instance, public punishments often enhance cooperation (Xiao and Houser, 2011). In general, emotions modulate our motivations and shape our decisions.

Psychophysiology measures external bodily states that are known to be closely linked to internal emotional and mental states, and generally linked to behavior. These measures usually have low marginal cost per person. But preprocessing and linking measures to behavior is usually not simple.

First, note that emotional reactions have two dimensions—"arousal" or intensity (the amount), and "valence", which refers to whether the emotion is pleasant or unpleasant.

Measuring arousal

Emotional arousal can be detected by measuring skin conductance response (SCR) at the time of an event. SCR has been widely used in psychology and neuroscience, but rarely in economics. This body of research shows a link between SCR and behavior, and SCR is regarded as a reliable indicator of general emotional arousal (Figner and Murphy, 2011; Boucsein, 2012; Dawson et al., 2017).

SCR can be measured using physiological systems like those developed by Biopac, or portable wrist-worn devices such as Embrace. Specifically, SCR assesses the electrical conductance of the skin, which varies with minor changes in moisture caused by physiological arousal. A spike in SCR signals arousal in the subject, usually with a 1-2 second time lag. It is crucial to time experimental events with sufficient inter-trial time intervals (ITIs)—typically around 7 seconds apart—to allow skin conductance levels to return to baseline (Figner and Murphy, 2011). This will usually slow down the pace of an experiment, compared to one without SCR measures, so an experimenter needs to be sensitive to the fact that subjects may become more bored or restless than usual.

More recently, portable wrist-worn devices (Empatica) and even rings (Oura) can also measure psychophysiology. Their portability means emotions can be measured in almost any outside-the-lab everyday or artificial situation (e.g., experiencing fear in a haunted house (Tashjian et al., 2022)).

Data Analysis

Skin conductance data are analyzed using tools like SCRalyze (Bach et al., 2010; Bach, 2014). Events that create arousal include decision events and feedback events. Arousal during a decision refers to the SCR recorded while the subject is deliberating. This is usually modeled as a "flexible duration" event, starting when the choice appears on the screen and ending when the individual makes a selection. Arousal during feedback refers to the SCR measured when the individual observes the result of their choice. This is modeled as a "fixed" event with no specified duration (a spike regressor), as the psychophysiological response is captured immediately after the feedback is received.

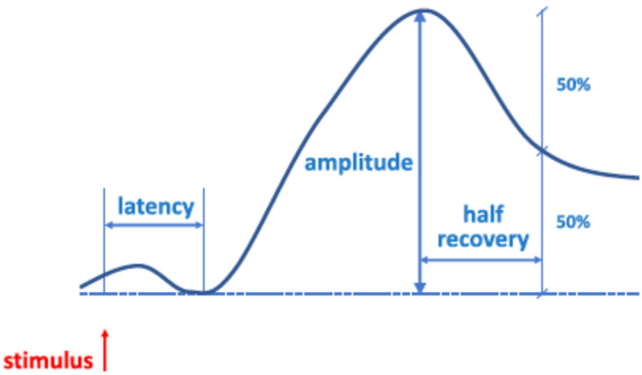

FIGURE 7 Representation of key physiological measures from SCR data (Source: Brocas et al., 2021).

Measures

For flexible duration events, the software estimates three physiological measures: Latency, amplitude, and half-recovery time (HRT) (Dawson et al., 2017). They are illustrated in Fig. 7. Latency is the time interval between the stimulus onset and the initiation of the SCR, measured in seconds. Amplitude is the maximum increase in conductance after the end of the latency period, measured in microsiemens. Half-recovery time (HRT) is the time interval in seconds between the SCR peak and the point where it falls halfway back from the amplitude peak to the post-latency baseline from the peak. On each trial, these three numbers measure how long it takes to respond to the stimulus (latency), the level of arousal (amplitude), and the duration of the arousal (HRT).

For fixed events, the software only estimates amplitude and assumes canonical parameters for latency and HRT.

Non-specific SCR data (as opposed to event-related SCRs) are used to compute a Lability Index. This index is obtained by counting the number of SCRs over a fixed period, typically a few minutes before the experiment starts (Brocas et al., 2021). The electrodermal response lability index is relatively consistent within individuals and is associated with certain personality traits. Specifically, individuals with more SCRs ("labiles") tend to be emotionally undemonstrative, introverted, agreeable, and calm. Conversely, those with fewer SCRs ("stabiles") tend to be emotionally expressive, extroverted, impulsive, and antagonistic (Crider, 2008).

Brocas et al. (2021) measure SCR in a steal-or-pay game. One player exerts effort to create a surplus of $20. A "dictator" can either pay $7 (earning $13 net) or just steal the $20. In a Part I condition, there is a $1 fine if the stealer is caught (40% probability) but the stealer still keeps the good, therefore obtaining a minimum net utility of $19. In a subsequent Part II condition, in addition to a fine, a picture of the thief is displayed to subjects and they are described as a cheater. Arousal was significant at the time of decision but it was modest at

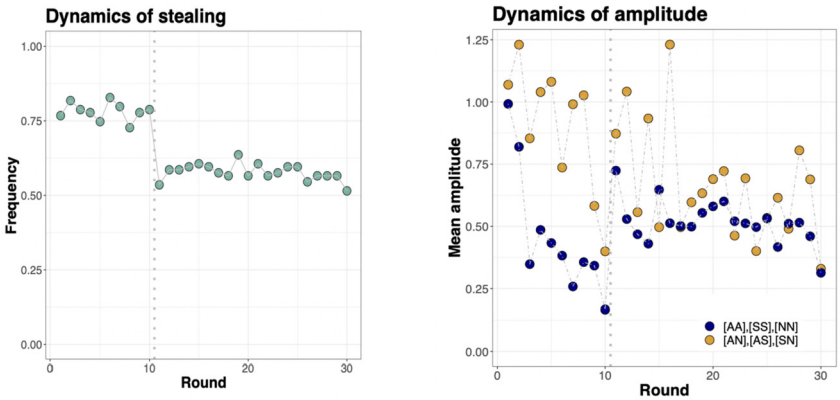

FIGURE 8 *Left*: aggregate stealing behavior in Part I (round 1 to 10) and Part II (rounds 11 to 30). *Right*: Amplitude of SCR at decision. Blue dots (low mean amplitude in Part I) are individuals who did not change behavior between the two parts, while orange dots (low mean amplitude in Part II represent those who significantly reduced their stealing tendency). Source: Brocas et al. (2021).

feedback in both parts. Importantly, anticipated shame reduced stealing from 75% to 60%. Arousal exhibited at the time of the steal decision in Part I was also associated with switching decisions from Part I to Part II (Fig. 8).

A second experimental economics example uses SCR to measure anxiety in dynamic and static games. Kang and Camerer (2018) recorded SCR during a stylized "clock game", which was cleverly designed to illustrate how stock market bubbles might arise (Brunnermeier and Morgan, 2010).

In the clock game, a stock price is rising exponentially based on some reduced-form exogenous process. At some point, the rising price exceeds a fundamental value. Participants know the price at any time, but not the hidden fundamental. In a window of time after that crossing point, each of six subjects get independent announcements that the price is higher than the fundamental. They have to decide when to sell—at the exogeneous increasing price—or whether to hang on longer. Once three of the six subjects sell, the price crashes to the fundamental value.[5]

The clock game is carefully designed so that there is an optimal time to delay selling after receiving a signal. Brunnermeier and Morgan (2010) found that experimental subjects sold sooner than theory predicted. Kang and Camerer (2018) hypothesized that anxiety might explain the early selling. They found that people sold later in a static experimental condition in which they know when their signal arrives, and have to state when to sell without watching prices rise (like a sealed-bid auction compared to English or Dutch dynamic-price change auctions; also called the "strategy method'' in experimental economics).

Their measured SCR's are consistent with the idea that anxiety is a negative emotion and creates anticipatory disutility, which motivated early selling in the

[5] The clock game is similar to a multiple-person "balloon" BART design used to measure risk-aversion (Lejuez et al., 2002).

early dynamic treatments (a la Loewenstein, 1987; Caplin and Leahy, 2001). SCR is higher in the dynamic than static case. Within subjects, differences in SCR in the two dynamic and static conditions are also strongly correlated with delay differences (r = 0.68). Their conclusion is that to fully understand what is happening in this elegant game requires some understanding of emotions such as anxiety, and those emotions can be measured by SCR.[6]

Besides the stealing and clock game examples, SCR arousal has also been associated with immoral behavior in economic experiments on tax evasion (Coricelli et al., 2010) and both free riding, and others' freeriding, in public goods games (Joffily et al., 2014).

Most devices that measure SCR also measure heart rate with ECG (heart rate variability too) and other biomarkers such as temperature. Bossaerts et al. (2024) collected these measures in a declining-dividend asset market where assets are typically mispriced in bubbles (deviations from intrinsic fundamental value). They found that a trader's heart rate *anticipation* of their bubble-price trades was predictive of higher earnings, and the heart rate *reaction* to trading had the opposite effect (lower earnings)

Facial electromyography (EMG) is a more difficult method to measure emotions. EMG records changes in facial muscles, which are known to be associated with distinct emotions. Basic emotions are associated with small changes in muscles that can be recorded by electrodes fixed to facial areas.

For example, when people are disgusted, the levator labii muscles raise the upper lip and wrinkle the nose. Chapman et al. (2009) recorded facial EMG as people tasted neutral, sweet or bitter drinks and found strong levator labii reaction to bitter taste. They then had subjects play an ultimatum game in which they got $5, 3, 2 or 1 offers out of $10. The same levator labii muscles which reacted to bitter taste also reacted more strongly to $1-2 offers (more than the other offers). The strength of the EMG reaction was correlated r = 0.70 with rejecting offers.[7]

3.2 Measuring brain activity

Direct measurement of brain activity is also process data. We describe some examples and considerations only briefly here. The brevity is because it is much harder for Handbook readers to use these methods effectively without special training (e.g., a postdoctoral experience) or as part of a scientific team with

[6] Kang and Camerer (2018) was difficult to publish in a selective economics journal. Referee reports from a top 4 journal indicated the range of acceptance of data such as SCR. A skeptic wrote "I am not sure what an economics researcher learns from this paper... All it does is show a correlation between the difference in behavior and some physiological measure that another discipline associates with the term "anxiety"." Another referee was more positive. They wrote "This is possibly the most important experimental paper ever written connecting the emotions with decision making."

[7] An important early neuroeconomics paper (Sanfey et al., 2003) found that insula activity was associated with ultimatum rejections. This is consistent with the EMG results because the insula is reliably activated by unpleasant bodily mental experiences which evoke disgust.

neuroscience experts. There are also many other sources of information about how exactly to use these methods and what has been learned (Glimcher, 2013).

There are a few different methods to measure brain activity "non-invasively" (i.e., without putting electrodes into brain tissue). We will talk only about functional magnetic resonance imaging (fMRI) and electroencephalography (EEG) though there are other less frequently used methods such as positron emission tomography (PET), functional near infrared spectroscopy (fNIRS), and magnetoencephalography (MEG).

To measure where activity in the brain is occurring, the most popular technique is fMRI. fMRI measures the ratio of oxygenated to deoxygenated blood at each point (3 mm cubes that are called voxels) in the brain. This "BOLD" signal is a proxy for brain activity, which requires oxygen. Sectioning the brain into 3 mm cubes yields approximately 100,000 voxels, and recording from all these voxels takes approximately 2 seconds (though this time is shrinking as technology improves). However, a fundamental limitation of this technique is that the BOLD signal is slow—it peaks 4–8 seconds after the corresponding brain activity and takes 10 seconds to return to baseline. Thus, fMRI is useful for imaging the whole brain and identifying the location of activity, but less useful for studying the dynamics of the activity.

EEG measures electrical signals on the scalp due to underlying neural activity near the surface of the brain. EEG has excellent time resolution, on the order of milliseconds, but it has poor spatial resolution—it only accurately measures activity near the cortical surface. Much as an array of earthquake seismometer measures can be used to pinpoint the source of a quake, multiple EEG signals can sometimes be used to guess at the regional source below the cortical surface.

fMRI and EEG are very complementary techniques. fMRI excels in the spatial dimension while EEG excels in the time dimension. Some recent work has sought to combine the advantages of each technique by simultaneously recording fMRI and EEG data. While extremely technically challenging, this research has shed important light onto the choice process (Pisauro et al., 2017).

An advantage of neuroscience is bio-informatic accumulation of knowledge: Every modern fMRI study reports voxel-level activity using a conventional (x,y,z) coordinate system (right-left, back-front, bottom-top aka right-left, posterior-anterior, ventral-dorsal). A website called Neurosynth (https://neurosynth.org/) has compiled more than 10,000 studies. It automated extraction of both voxel (x,y,z) locations of peak activity during a task, and verbal labels of what authors think that task activity is identifying functionally. (We'll see an example below.)

Neuroeconomists are always striving to answer the question: "Why care about where?" That is, what is learned by knowing that putting a larger weight on future rewards (as inferred by an increase in the discount factor) is associated with hippocampal-cingulate coactivation, rather than by dLPFC, frontal pole, or some other region?

The main general reason is that knowing the regions of circuitry (the "where") often enables us to make guesses about what mental computation is

being done, if we have sufficient information from many other studies about the neural functions of the regions that were identified.

Here is an illustrative example from analysis of experimental stock price bubbles. There are many, many theories in financial economics of why bubbles form and crash. Many of these theories depend on institutional market details (e.g. lock-up periods for insider trading of tech stocks, asymmetries in private information, valuation uncertainty combined with short-selling restrictions, etc.). One might say that these theories lack 'external validity' (aka generalizability) because they do not apply to a wide range of assets, markets, and historical periods.

When theories depend on institutional features, their theoretical predictions can also be disabled in simpler experimental environments which do not have those institutional features. To this end, there are about a hundred artificial stock market experiments, mostly adopting and modifying the Smith et al. (1988) design in which the fundamental value is declining across trading periods (Palan, 2013).

Smith et al. (2014) used a different design in which the hypothesized equilibrium asset value is constant (Holt et al., 2017). In their design, assets pay an expected dividend of .7 units per period, but cash holdings can also earn a risk-free return of 5%. Assets were redeemed at a known value of 14 after the final trading period 50. This design creates an equilibrium condition (Euler equation): asset prices should satisfy the equation $.05P^* = .7$ which implies $P^* = 14$.

Smith et al. found frequent bubbles in which the price rose above the predicted fundamental of $P^* = 14$. The mean bubble peak was about 40, and it peaked around period 35 of 50. Using fMRI, Smith et al. (2014) found that people who were selling near the height of a bubble had increased activity in the insula, and people who were buying had slightly decreased activity in that region (and increased activity in nucleus accumbens, a reward-tracking region). Many studies have found that insula is persistently activated by pain, financial uncertainty, task switching, and other functions. Some of these functions are consistent with the hypothesis that insula activity is an internal warning sign that at a certain time, trading in a bubble stock is more risky, and it's time to sell. Those who sold had a stronger neural warning; those who kept buying had a weaker warning from their insula.

Like many early findings in cognitive neuroscience, the interpretation just offered is essentially a new hypothesis based on discovery of insula activity using fMRI. There were dozens of previous economics experiments about stock price bubble formation. None of them derived this testable hypothesis about a special role of a certain kind of measurable emotional signal, identified from its "where" location and from previous findings about functionality of that same location. In one theory based on partial sophistication, the authors describe bubble trading as transitioning from "euphoria" to "panic" (Bianchi and Jehiel, 2010). fMRI neuroeconomics takes this language seriously as describing candidate mental states and tries to measure them directly.

Here is another example of clear economic interest. It is no secret that human choices are typically stochastic. There is intense interest in modern behavioral economics in whether stochasticity is due to random utility, to imperception in either perceptual encoding or response output, or some combination.

Polanía et al. (2014) contributed to this scientific conversation by using EEG to measure brain activity in both perceptual and economic decision making about foods—subjects were asked to choose either the larger food or their preferred food. They identified separate EEG signals for both types of judgments. When synchrony of these two signals (frontal and parietal) was higher, subjects' preferences were more internally consistent. This is an intriguing piece of evidence that a source of stochasticity is slippage between what people perceive and what they value. It is hard to see how you could make this inference conclusively without measuring brain activity during matched perceptual and economic choice tasks.

Furthermore, the EEG analysis clearly answers the "Why care about where?" question. In a follow-up study, Polanía et al. (2015) applied transcranial alternating-current stimulation (tACS) to specifically modulate communication between the frontal and parietal regions discovered with EEG, and found that this causally reduced subjects' economic (but not perceptual) choice consistency. Polania and his collaborators would not have known what signals to disrupt without the "Where?" information from EEG.

A final brief example is neuroforecasting. Understanding the causal relationship between attention and choice can inform neuroforecasting models to predict market trends or consumer preferences based on physiological and behavioral cues (Knutson and Genevsky, 2018).

For instance Falk et al. (2013) examines how neural responses predict the spread of ideas during social interactions. Using fMRI, researchers found that successful ideas activated mentalizing and reward systems in communicators' brains when first exposed to the messages. Similarly, Doré et al. (2019) explores how neural activity in the ventromedial prefrontal cortex (vmPFC) predicts the widespread sharing of information. Researchers found that individuals whose vmPFC activity closely tracked population-level sharing also aligned with social consensus about article value. Frequent news readers showed consistently high vmPFC activity, while infrequent readers responded strongly only to widely shared articles. These findings are critical to understand how information spreads in communities.

4 Causal manipulation of choice processes

A benefit of choice process data is that the association of choice process data and behavior almost always suggests hypotheses about potential causal effects of exogenously manipulating the choice process. And the technology used to measure the process variable often suggests precisely how to exogenously change that variable.

Causal manipulations can be implemented in various ways. We'll mention examples from manipulation of eyegaze, brain stimulation, induction of stress, and psychopharmacology.

4.1 Gaze and other process manipulations

We know that the amount of visual attention paid to a choice object in a binary choice is associated with actually choosing it (Krajbich, 2019). This process-data association suggests that exogenously increasing attention to a choice object might causally influence people to choose it more often.

Note well that this suggested hypothesis is *not* due to the common amateur confusion that correlation implies causation. The idea here is different: The overarching hypothesis is that "upstream" processes of deliberate attention allocation, and exogenous attention, are entangled (in the sense of Bayesian signal-extraction models; e.g. Cunningham and De Quidt, 2022). If the downstream system does not sufficiently signal-extract the influence of exogenous attention, then exogenous attention will have a causal effect.

And indeed, this relationship is causal: In a meta-analysis, Bhatnagar and Orquin (2022) show that increasing gaze on an option causes a small but reliable increase in choice probability in binary choice, from 50% to about 54%. Their meta-analysis reveals that dwell-time manipulations are effective, but manipulating the first-fixation doesn't seem to be as effective.

A related way to manipulate choice is to change the order in which attributes are considered. We know that there is a latency (slowdown) with which attributes enter the decision process (Sullivan et al., 2015; Maier et al., 2020; Chen et al., 2024a; Yang and Krajbich, 2023). A recent paper built on that idea and manipulated the order in which attributes were revealed to decision-makers. In an intertemporal choice task, subjects saw the rewards before the delays, or vice-versa. Subjects who saw the rewards first were ∼10% more likely to choose the larger-later option than subjects who saw the delays first (Chen et al., 2024b). Kwak and Huettel (2018) found a similar ordering of about 6% on gain-loss framing. A related order effect has been shown in "query theory". Asking people to verbally describe the reasons to sell an object first, compared to listing reasons to buy first, can erase and reverse endowment effects (Johnson et al., 2007). In general there seems to be a "primacy effect" in which the information brought to mind earliest is weighted more heavily than information that comes later.

Causal non-choice data methods offer valuable insights into decision-making processes and have several promising applications across domains. These methods make it possible to explore heterogeneity in decision-making by identifying how different individuals respond to process manipulations. This can help tailor interventions or nudges to specific subgroups for more effective outcomes.

Another kind of causal gaze manipulation is something most of us had never heard of before finishing this chapter: It's called "eye movement modeling examples" (EMME). The first example of this method is a remarkable experiment by Thomas and Lleras (2007).

They studied a famous "aha" insight problem called the Duncker radiation problem. In these insight or "epiphany" problems[8] (see Chen and Krajbich, 2017), people have little sense of conceptual progress until there is a sudden onset of confidence when they have found the right answer. The Duncker problem shows a small tumor in the center of a patch of skin. The challenge is that too much radiation aimed at the tumor will destroy the tumor but also harm the patient greatly. Is there a better way to destroy the tumor? The insight answer is to aim smaller doses of radiation from different angles around the skin patch, like soldiers surrounding an enemy.[9]

Thomas and Lleras manipulated gaze by having subjects fixate at a sequence of eight locations, which guided their eyes as if they were aiming from ideal different angles. Success rates increased from 20% to 50%. Importantly, few subjects self-reported an influence of the guided fixations on getting the answer right. The effect is implicit, but not explicitly recognized.

Since then, a standard paradigm for EMME in medicine is to show medical students eyetracking videos or heatmaps of doctors looking at, for example, radiographs searching for potential tumor locations. This has proven effective in medical training (Ashraf et al., 2018).

4.2 Transcranial electric/magnetic stimulation

Because neurons communicate via electric signals, brain activity can be manipulated using electric and magnetic fields which are "non-invasive" (meaning that electrodes are not inserted into brain tissue, as in single-unit and ECoG recording). Transcranial direct current stimulation (tDCS) and transcranial magnetic stimulation (TMS) are methods that neuroscientists use to alter brain activity. TMS causally inhibits neural firing and is thought to reduce activity in a targeted area. tDCS raises *or* lowers the threshold for a neuron to fire. The fact that tDCS can both upregulate (increase) and downregulate (decrease) activity by reversing the polarity of two target sites is a big advantage for causal exploration. TMS is a more powerful technique than tDCS but is also more expensive and requires more care to use (see Chapter 6 in Glimcher, 2013 and Ruff and Huettel, 2014 for more details).

Being able to manipulate brain activity exogenously is obviously important for establishing the causal role of brain activity in choice. In many cases, it is also used as a therapeutic treatment (e.g. TMS is used to treat major depressive disorder). Here are some examples illustrating the power of these techniques (beyond those in Table 1).

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[8] See Chen and Krajbich (2017) for a compelling example of identifying epiphany learning using eye-tracking and DDM. There are so many potential examples in economic decision making to cases where there is an optimal behavior that is suddenly, rather than gradually, discovered, ranging from learning to bid one's value in a second-price auction to R&D innovations.

[9] Then the radiation from each angle is low enough to not harm the skin, but all the radiation from the different angles arrives together at the tumor and kills it.

Krajbich et al. (2021) disrupted activity in the right frontal eye field, a brain region known for guiding visual attention. This disruption of attention reduced the effect of gaze on choice and increased RT, specifically for high-value decisions (as was uniquely predicted by their model).

Camus et al. (2009) disrupted activity in right DLPFC and reduced monetary valuations for foods; this shows that rDLPFC (part of an executive function/self control/working memory circuit) causally contributes to valuation.

Another example is Figner et al.'s (2010) use of TMS to downregulate rDLPFC (identified by Hare et al., 2008) which caused people to discount future rewards more heavily.

A HD-tDCS example is Li et al. (2020). The tDCS method often generates small effects. However, it has the advantage that two electrodes are placed on the skull, and reversing their polarity changes whether activity is increased or "upregulated" (anodal condition) or decreased or "downregulated" (cathodal condition). The TPJ is a region that is well established as a part of a "theory of mind" mentalizing circuitry. They found that upregulation of that region made people more generous in allocating money in dictator games, and also enhanced the consistency of their decisions (as measured by conventional rationality indices such as Afriat).

4.3 Stress induction

Stress is a physiological and psychological response to perceived challenges or threats ("stressors"). Under stress, the body activates the hypothalamic-pituitary-adrenal (HPA) axis, leading to the release of cortisol, a reliable biomarker of stress. Cortisol significantly influences decision-making processes. Elevated cortisol levels can impact executive function in the prefrontal cortex.

Under acute (temporary) stress, cortisol enhances focus and prompts quick, survival-oriented decisions. However, chronic stress, causing sustained high cortisol levels, can impair cognitive functions, leading to difficulties in concentration, memory, and logical reasoning. The quality of decision-making usually suffers.

Biological stress is often measured from saliva samples which can detect cortisol, using cotton swabs or "drool" techniques. *Creating* stress exogenously and reliably across people is remarkably easy. One method is the Socially Evaluated Cold Pressor Test (SECPT). This measure combines physical and social stressors by having participants immerse their hand in ice-cold water (0–4°C/32–39°F) while being video recorded or evaluated, mimicking natural stress scenarios like public speaking. The task induces a physiological stress response and enhances ecological validity through social observation. A between-subjects design is typically used to compare stress and control groups, minimizing adaptation effects. The cold water method is slightly painful but compliance is typically high. One added benefit is that, despite the very large increase in cortisol level, participants in the Cold Pressor Task do not feel higher levels of

stress in post-task surveys compared to control individuals, which eliminates explanations based on demand effects.

The Yerkes-Dodson Law (Yerkes and Dodson, 1908) posits that stress levels above an optimal threshold unambiguously impair performance; this is called "choking" in sports and is also important in children with test-taking anxiety (see Chib et al., 2014 for neural evidence and a connection to loss-aversion, and Dunne et al., 2019 for methods to reduce choking). In some cases, novel high incentive tasks actually reduce performance. This hypothesis is a challenge to theories of rational inattention which predict no such effect.

The empirical evidence about stress and aspects of decision-making is mixed. Mather and Lighthall (2012) and Starcke and Brand (2012) review evidence about stress and risk preference. Studies using incentivized lotteries indicate that stressed males tend to choose riskier options, whereas stressed females choose less risky options (Preston et al., 2007; Lighthall et al., 2009; Van den Bos et al., 2009). Additionally, chronic stress over an eight-day period has been shown to significantly increase risk aversion compared to acute stress (Kandasamy et al., 2014). Research by Brocas et al. (2017) demonstrated that the behavior of stressed individuals in lotteries, Stag Hunt, and Hawk-Dove games is highly correlated across those three contextual domains. This strong relationship was not observed in the unstressed control group, highlighting a phenomenon termed "contextual blindness" induced by stress. One possible explanation for the significant connection across tasks under stress is that stressed individuals tend to rely on a heuristic mechanism for decision-making across various contexts, while unstressed individuals adopt a more cognitively flexible approach. These findings are based on manipulations designed to establish a causal link between stress and behavior.

4.4 Pharmacological treatment

Many studies in clinical and cognitive psychology, and a few with economic games, have explored whether administering pharmacological[10] substances causally changes behavior. Note that for most experimental economists, conducting such studies requires collaboration with domain experts (and a reasonable IRB), or with particularly adventurous PhD students (as in Nave et al., 2017, which included an expert on measuring testosterone from saliva samples). We'll discuss three briefly: Oxytocin (OT), arginine vasopressin (AVP), and testosterone (T).

OT: Oxytocin is a neuropeptide, a "chemical messenger" in the brain. In humans, OT is important during lactation and childbirth, and also during social attachment. Kosfeld et al. (2005) created a stir when they reported that exogenously administering OT increased trust in a trust game. That intriguing finding was cited 2400 times (as of Dec 2024, Web of Science) but unfortunately has

[10] The term "psychopharmacology" refers to the study of how administering psychoactive substances might change psychological processes and subsequent behavior.

not held up well, in a highly-powered replication study (Declerck et al., 2020). The causal effects of OT are mixed and hard to pin down (Nave et al., 2015). OT is particularly challenging to study in humans because it does not cross the blood-brain barrier, which prevents large molecules (such as OT) from directly entering the brain.[11]

While the early 2005 results showing that OT increases trust did not replicate, there is still a lot of promising research. Andari et al. (2010), for instance, found that administering OT to autistic adults enhanced their trust in prosocial partners and attention to the eyes in a face (which autists generally look at less than neurotypicals).

AVP: AVP is also a neuropeptide which has a secondary role in creating social affiliation. Administering AVP to rodent montane voles, which are normally promiscuous, makes them 'good husbands'—they have an increased preference for familiar female sexual partners and are more aggressive toward stranger males (Young et al., 1998).

An experimental advantage of AVP, compared to OT, is that it crosses the blood-brain barrier. Brunnlieb et al. (2016) administered AVP to human subjects when they played versions of a 2x2 "stag hunt" game. Stag hunt has two pure Nash equilibria, which are Pareto-ranked; but choosing the "stag" is riskier because the payoff is low if the other player does not choose stag too.

They found that AVP increased the willingness to engage in the risky but socially beneficial stag choice from 40% to 50%. This increase was not attributable to a greater general willingness to bear risks or to help others, nor did AVP alter beliefs about the choices of others. fMRI also showed that the causal AVP increased connectivity between DLPFC (a cognitive control region) and the ventral pallidum region of the brain, which is known to have many AVP receptors.

T: Testosterone is a sex hormone, present at higher levels in men, which regulates physical growth processes and sex drive. In humans, a prominent idea is that T also promotes aggression when status is challenged (that's the "challenge hypothesis").

Prenatal exposure to T is associated with a simple measure of the ratio of the length of the index finger to the ring finger (called the "2D:4D" ratio) after birth. This ratio has been studied a lot and correlated with risk-aversion, etc. but the results are mixed.

Many studies have causally administered T exogenously to test its effects on risk-aversion and other behaviors. The results are also mixed. One interesting result for economists is that a dose of T seems to increase preference for status goods (Nave et al., 2018), which is consistent with a modern interpretation of the challenge hypothesis, if status goods are like cultural 'weapons of aggression'.

Despite the fascination with causal effects of psychopharmacology (particularly the OT-trust link), it is hard work and requires great care and large samples. The effects are rarely straightforward.

[11] A major challenge in OT research is that OT is usually exogenously administered nasally, through an inhaler, but only a small percentage crosses the blood-brain barrier and enters the brain. It is therefore an open question of how causal administration of OT actually works (e.g., Yamamoto and Higashida, 2020).

5 Conclusions and looking forward

This chapter discusses how and why to measure choice processes in economics experiments.

First, note that some important methods were omitted for brevity. These include:

- Virtual reality. This can allow researchers to create highly realistic and believable experiences, bringing the field into the lab (Noussair et al., 2024).
- Patient populations. These allow researchers to study deficits in decision making due to specific brain damage or psychological disorders. They can be akin to quasi-causal experiments on humans which are not otherwise ethical.
- Animals. Allows researchers to study how economic behavior has evolved across species, and allows for a broader range of neural measurement and manipulation techniques which can be rarely or never used on humans.
- Genetics. Allows researchers to study the role of nature vs. nurture, and gene-environment interactions, in economic behavior.
- Children. Allows researchers to study the development of economic behavior ("ontogeny").

Experiments with children deserve special mention because human development ("ontogeny") starting at young ages can have such permanent and impactful effects on preferences, human capital formation, and life outcomes (e.g. List et al., 2023). Many experimental studies explore the mechanisms underlying economic decision-making to examine what children and adolescents can grasp early, what abilities develop gradually, and what skills may never fully mature. For example, Brocas and Carrillo (2021a) investigate market exchanges—one of the most fundamental economic environments—and find that children as young as five years old understand how to use commodities as a medium of exchange. In contrast, Brocas and Carrillo (2022) study the use of mixed strategies in strategic settings and observe minimal development from kindergarten through adulthood. Positioned between these outcomes, Brocas and Carrillo (2021b) analyze the progression of strategic reasoning in children and adolescents through games involving iterative thinking, showing that these skills improve during elementary school years but tend to plateau by middle school. These patterns reflect biological changes that shape cognitive and socio-emotional development, which in turn influence decision-making processes.

The main themes discussed in the chapter are the following:

- There are many measurable choice processes going on between the time the experimenter poses a question and the time the participant makes a choice. Not recording process data is often a missed opportunity to understand something fundamental and diagnostic (e.g., for distinguishing between theories).
- Some measures are easy to collect and analyze—RTs and mousetracked trajectories. They provide reliable information about strength-of-preference. There are few experiments in which these data would not be of some use.

- Other measures require more software or machinery, such as Mouselab and eye-tracking. Think of these as unpacking what information is being attended to during the decision. These have been useful, historically, in experimental economics in understanding multi-attribute choice, the heuristics people use in response to choice complexity, and the strategies people use in games. An aggressive view is that any study which is about salience or attention can find these measures to be useful as validation of those basic constructs. For example, suppose a theory claims feature X is salient and infers so from choice, but X is not looked at early or often. This would call into question the idea that X is important because it is salient. If inferred salience and measured attention do not match up there is important work to do in reconciling the inference and the measure.
- Psychophysiology includes measures of skin conductance, heart rate, and other bodily measures often associated with mood and emotion. Highly portable sensors are now available to provide these measures both in-lab and in everyday settings, with millisecond time resolution. Fixed costs are modest ($.5-2k/person) and marginal costs are near zero.
- We briefly discussed more complicated methods such as EEG, fMRI and brain stimulation. We are big fans and users of these methods and advocate for more widespread use in experimental economics. However, setup costs in learning how to use them and marginal costs per subject are much higher.

Readers of this chapter are likely to have two distinct questions:

- Technical questions: How expensive are these methods? How hard are the methods to learn? What are successful examples of using process data to answer fundamental questions? We have given some information to help answer those very sensible questions.
- The harder question is "Can I publish papers in economics using these methods?" The answer will constantly be in flux because it will depend on the tastes of referees and editors and perhaps recent experience. It is certainly true that it has *never* been easy to publish papers in economics using process data (or, for that matter, any other nontraditional methodology). The prevailing view among referees and editors, in our substantial experience, is that theories can usually be adequately tested by choice data alone (with beliefs and surveys being more accepted additions). Process data are generally regarded as supportive, but not definitive. Unfortunately, it is even difficult to have much public discussion of this implicit, decentralized editorial choice. On the other hand, originality may pay off in the long run.

We have opted to classify methods into three categories: simple, complex, and causal. While this classification is somewhat arbitrary, it serves to organize the approaches effectively. **Simple non-choice data methods** aim to describe observable non-choice mechanisms that precede or correlate with choice. **Complex methods** delve deeper, focusing on biological processes that underlie observable choice and non-choice outcomes. In contrast, **causal methods** are designed to manipulate these underlying processes directly.

For example, longer RT during decision-making might indicate difficulty in comparing options that are too similar or in retrieving information necessary to differentiate two options that are ultimately judged as distinct. Incorporating brain imaging into this paradigm could reveal whether the decision process primarily engages brain systems related to attention or memory. Meanwhile, employing TMS could modulate activity in the working memory system, thereby altering RT. Each method provides unique insights into the mechanisms driving choices, offering varying levels of granularity.

Non-choice data methods can be used to address three key questions of importance to economic science.

- Non-choice data help evaluate the validity of economic theories and their ability to accurately capture the properties of decision-making. Traditional economic models rely on an "as if" approach, simplifying complex decision mechanisms into tractable mathematical representations (e.g., utility functions to model valuation processes in the brain) while assuming decision-makers possess infinite attentional, computational, and logical resources. Non-choice data challenge these assumptions, revealing for instance that utilities are not fixed entities; instead, values are computed dynamically with each decision, and the valuation can vary across episodes. Additionally, non-choice data show that the brain may not implement the algorithms necessary to achieve best-response strategies in dominance-solvable games. In some cases, the discrepancies between the predictions of economic models and observed behavior are substantial. Non-choice data can pinpoint where these models fall short and guide their refinement to improve predictive accuracy.
- Non-choice data address methodological considerations in decision-making research. For instance, choice outcomes and RT are both critical to understanding decisions, as they complement one another in revealing the motivations behind choices. Methodologically, RTs serve as proxies for the complex cognitive processes that occur before and during choice, making them a valuable source of information to collect alongside choice data.
- Non-choice data illuminate heterogeneity in behavior and its underlying causes. Behavioral variability is often treated as a "black box," but non-choice data show that parameters exhibiting variation frequently correlate with other factors that share similar patterns. For example, IQ has been linked to risk-taking and time-discounting behaviors, and these measures are independently associated with structural and functional properties of the same brain regions. This suggests that variability in one type of behavior can predict variability in another and may also provide insights into why these behaviors are interconnected.

5.1 Results of a *de facto* focus group

At an EER Experimental Economics Summer School in Rome (3 June 2025), our second author used the interactive app Slido to poll students about which

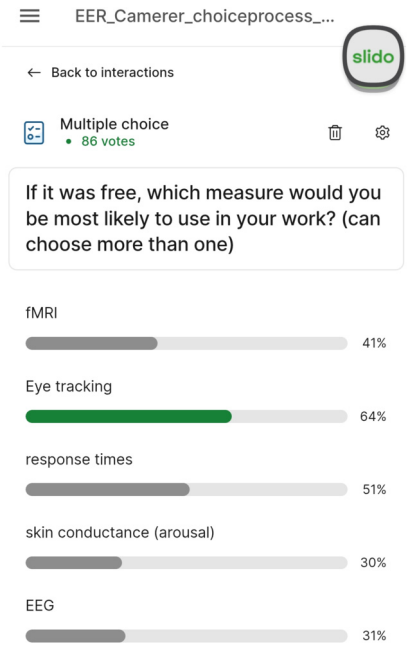

FIGURE 9 Results of a Slido audience poll. (Note that polled students can choose more than one item). Source: EER Summer School, 3 June 2025 LUISS Rome (organizer Michalis Drouvelis).

choice process measurement methods they might use (if the methods were free, liberating them from having to think too hard about cost-benefit calculations).

Fig. 9 shows the results. Respondents were most keen on eyetracking and response times. Note that the modest takeup rates (31–64%) imply that many respondents only chose one method. (Failure to include a "None" category also reduces what can be learned.)

5.2 Specific research topics

If you are a career-concerned experimental economist in 2025, you might be interested in a wide variety of topics and their economic implications. Here are some examples popular in c 2025 experimental behavioral economics in which we think process measures could be especially helpful (Table 3). Note, however, that the table is intended merely to inspire ideas about potential topics, non-choice data methods of interest, and possible testable hypotheses, which may not represent currently accepted interpretations.

Our shared view is that If you are investigating these topics, you will have to work hard to design an experiment to conclusively infer any of the latent variables using *only* choices. Survey responses, self-reports and RTs will certainly help. And it is generally even harder to design an experiment to distinguish *different* theories using only choice data.

TABLE 3 How choice process measures might help understand modern concepts in experimental economics.

	RT	Eye-tracking	Mouse-tracking/ mouselab	fMRI	Other
Complexity	higher	more fixations, more saccades and transitions	more nonlinear, curved trajectories reflecting hesitation and difficulty	Insula, ACC, more regions in MVPA (cognitive load and working memory demands)	
Procedural Rationality	Consistent RT across steps of reasoning		Predictable, linear movements	PFC	Voice: Even tone and structured speech
Salience	Faster responses	Increased fixation and pupil dilation		Salience network, AI, ACC, Amy, VS	SCR: higher arousal
Motivated reasoning	Longer RTs when priors are challenged	Increased fixation for confirmatory evidence		Self-serving bias/value processing in vmPFC and ACC	SCR: arousal when information is dissonant
Cognitive uncertainty	Longer times for increased uncertainty		Erratic, hesitant	conflict detection: ACC	SCR: more arousal when more uncertainty
Lying costs	Longer times to put effort to suppress truth	pupil dilation		cognitive control, conflict monitoring: PFC, ACC	SCR: increased arousal when preparing deceptive response; Voice: pitch changes
Levels of strategic thinking	Longer times for higher levels		Smooth movements for simple strategies, erratic paths for complex strategies	TPJ, medial PFC, temporal lobe	
Prosociality		Focus on others' outcomes		vMPFC, TPJ, DMN	Voice: soft tone

continued on next page

TABLE 3 (*continued*)

	RT	Eye-tracking	Mouse-tracking/ mouselab	fMRI	Other
Non-instrumental value of information	Fast RTs when high value	More fixation on high information, pupil dilation		value regions, vmpfc, VStr	
Discrimination (Implicit bias)	Longer RT when bias			Amygdala	Voice: hesitation when bias SCR: arousal for outgroup stimuli
Discrimination (perceptual, cognitive)	Faster RTs or easier discrimination	Longer fixation on difficult option to evaluate; saccades when indifference; pupil dilation if cognitive effort	straight paths for easy discrimination; delay starting movement and curved paths when difficult	Sensory cortices; complexity in dlpfc; conflict resolution when similar items in ACC; subjective value difference in vmpfc	SCR: higher arousal when difficulty; voice: speed and pitch
Mental Models		Focus	curved paths	abstract thinking: PFC	
Emotions		Pupil dilation for high emotional content		Depends on emotion	SCR: arousal for high emotional content; Voice: erratic, pitch, tone

Here is an example: In labor markets, taste discrimination is defined as being less likely to hire a person you dislike. Statistical discrimination means not hiring someone because they are a member of a group thought to be low-skill, and it is too costly to tell how different a particular candidate is from their (perceived-to-be) low-skill group members. These two causes of discrimination are not too easy to tell apart from choices alone.

Collecting attentional data from Mouselab-type lookups is a big help (Bartoš et al., 2016). However, more can be done. We conjecture that taste discrimination will also be accompanied—when looking at pictures of job candidates, for example—by negative emotions, insula or amygdala activity in the brain (Kubota et al., 2012; Kubota, 2024), other biomarkers, and IAT scores. Statistical and taste discrimination will be differently associated with cognitive uncertainty, which could possibly be measured by mouse-tracking, by ACC activity in the brain, and by pupil dilation. And note that people may make *implicitly* discriminatory judgments without *explicitly* acknowledging that. So self-reports cannot be solely relied upon, especially when other data are available.[12]

In closing, three final remarks are in order. One is that many studies which use process measures to study economic decisions are not published in familiar economic journals. For example, many of the examples in Table 1 are in general science (e.g., PNAS) or neuroscience journals. If experimental and behavioral economists want to cumulate regularity across *all* sciences, it behooves us to do broader literature reviews. Happily, the ability to do so using modern informatic search is easier than ever before.

The second remark is just a reminder that several examples in this paper show how choices can be *causally* changed via attention and other methods. Choices and RTs, food values,[13] patience toward future rewards, risk choices and honesty have all been shown to be causally changeable. Perhaps the most consequential example is evidence from tDCS in the TPJ brain region that rationality-0—as conventionally measured by consistency with utility maximization—can be increased. This type of causal effect goes to the very heart of what economists care about. In conventional theory, causality is presumed to work through prices, information, or constraints (e.g. taxes or subsidies). The causal examples described herein do not work through these through causal channels. They work by changing elements of the choice process.

The third remark, echoing our Simple, Complex, and Causal distinctions, is that the simple processes are all within the reach of most graduate students (except in-lab eye-tracking, which can be a little harder and more expensive in

[12] Indeed, the pioneering fMRI study by Phelps et al. (2000) found that differential amygdala activity of white subjects when viewing black vs white faces was uncorrelated across people with a standard Modern Racism Scale that uses self-reported attitudes. Neural activity *was* however, correlated with IAT scores and with the eyeblink startle response (a standard measure of fear).

[13] Ai et al. (2018) also increased snack food valuations by 8% using an amazing method called targeted memory reactivation (TMR). In their study, the name of a snack food was whispered when a person was in the N1 phase of sleep. Hearing information in this phase is known to increase memory.

fixed cost[14]). The others almost always require a collaboration with someone outside an economics department; but there is also likely to be *someone* at your university, or in your area, who teaches courses and owns equipment they might be interested in sharing.

Our general conclusion is that process data expands your toolkit, potentially helping you more efficiently, and conclusively, better understand many of the topics we are all trying to learn about in economics.

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[14] There are exceptions: Chan Lim bought a $5000 eyetracker himself and taught himself how to acquire and analyze data to study experimental subjects' expectation formation from looking at price charts (Gulen and Lim, 2024).

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